HANDBOOK

Laser Doppler Velocimetry

LDV system fp50 shift/unshift

and

software LDA Control QT

Intelligent Laser Applications GmbH

ILA GmbH
Karl-Heinz-Beckurts-Str. 13
D-52428 Jülich

Tel.: +49 2461 690-430
Fax: +49 2461 690-439
info@ila.de

www.ila.de

CONTENT

0. Safety Instructions

The laser optical measuring system works either with a HeNe laser (10 mW or 22 mW) or Nd:YAG lasers class 3B with a maximal power up to 500 mW.

LASER RADIATION
AVOID EXPOSURE TO BEAM
CLASS 3B LASER PRODUCT

Please avoid any reflection of the laser light on reflective or metallic surfaces. In any case, please cover these surfaces.

Don‘t stare into the beams coming out of the probe!!!

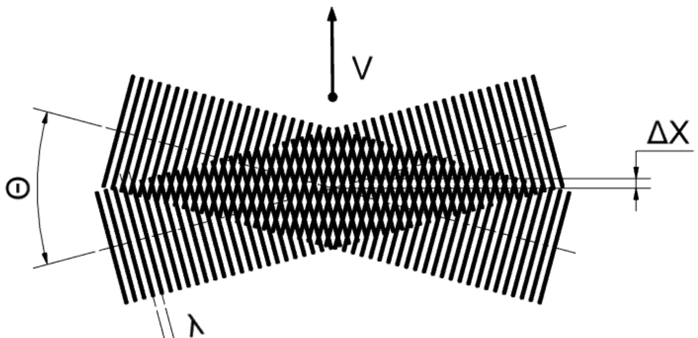
After using the measuring system with the HeNe-Laser, first switch off the laser and disconnect the controller and laser at least 15 minutes later, because the generated high voltage in the laser tube will disappear very slowly.

1. Measuring Principle

1.1 Unshifted Systems

The LDA control QT Laser velocimeter is a non-intrusive, highly accurate instrument for measuring flow velocities and is based on the principle of Laser Doppler Velocimetry. The optical accessibility of the flow is the main assumption for making measurements, as well as the existence of particles (0.5-100 µm) in the fluid.

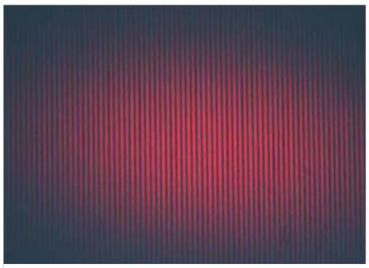
Two converging laser beams intersect each other, generating an overlapping region, which is known as measurement volume. A pattern of interference fringes is created in the measurement volume, where the distance between the interference fringes is defined by the wavelength of the laser light and the angle between the two laser beams, as depicted in Figure 1



$$\Delta X = \frac{\lambda}{2 \sin(\Theta/2)}$$

$$V = \Delta X * f_D$$

Figure 1: LDV Measuring Principle



The backscattered light from the MV is transformed into a voltage signal by a photodetector, then filtered and amplified in the controller.

Microscopic particles present in the flow backscatter the laser light at a frequency proportional to the flow velocity when passing through the measuring volume (MV).This back scattered light is transformed into a voltage signal by a photodetector, then filtered and amplified in the LDV controller. The Doppler frequency contained in this signal is determined by FFT processing.

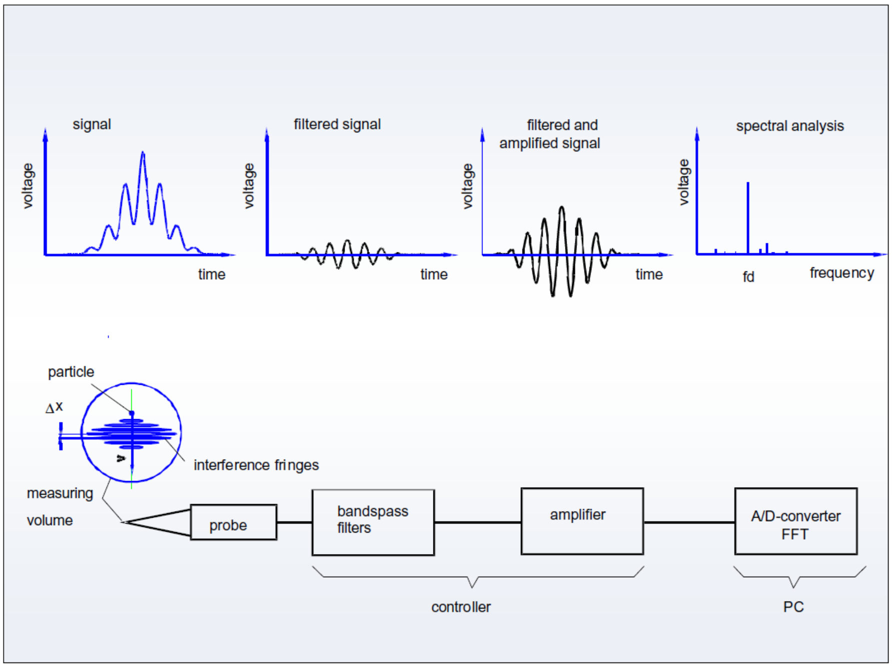


Figure 2: Signal processing

Figure 3 shows a sketch of principle of a combined sending and receiving LDV optics and the typical signal produced by a flow particle travelling through the measurement volume. The transient Doppler frequency is calculated by means of a special short spectral analysis. The particle velocity v is the product of the Doppler frequency f_D and the interference fringe distance Δx.

1.2 Shifted Systems

Detecting the direction of the flow requires the use of so-called “shifted” optics. Unshifted optics, where the interference fringes present in the measurement volume are static, cannot differentiate between particles entering the measurement volume from opposite directions, hence can provide the magnitude of the flow velocity but not its sign. By contrast, shifted optics allows the detection of flow direction by shifting the frequency of one of the Laser beams with an opto-acoustic device such as a Bragg cell (Figure 3)

.

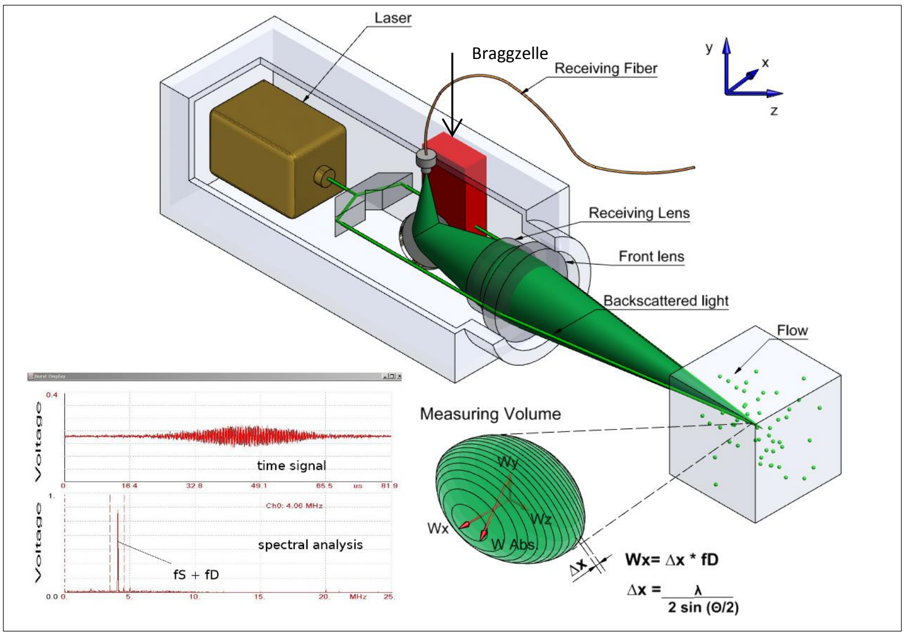
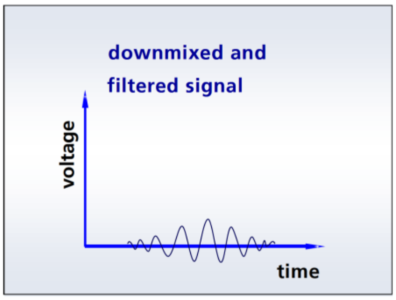


Figure 3: LDV shifted optic schematic and generated signal

The crossing of the shifted beam with the unshifted one results in a moving interference pattern inside the measuring volume. Driving the Bragg cell with at 40 MHz will cause the interference pattern to move at a 40 MHz frequency for a static observer. Particles moving in the direction opposite to that of the interference fringes will produce a signal frequency higher than 40 MHz (40 MHz + f_D). Particles moving in the same direction as the interference fringes will produce a signal frequency lower than 40 MHz (40 MHz - f_D). This frequency discrimination method allows the detection of the flow direction. To enhance the resolution of the FFT it is useful to downmix the Doppler signal frequency by means of a programmable synthesizer. This results in a lower sampling frequency and in a higher FFT resolution. See Figure 4 for further information.



Enhance the resolution of the FFT by downmixing the Doppler signal.

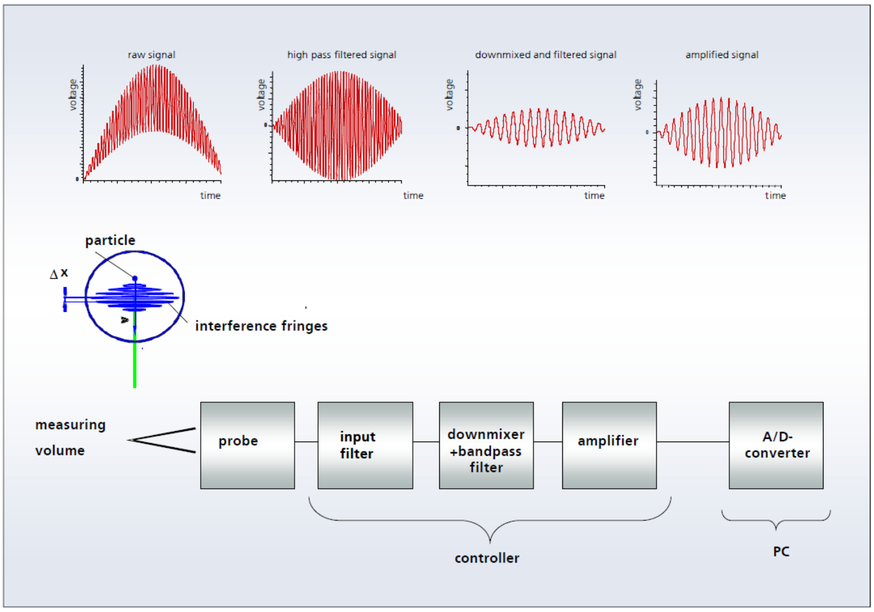


Figure 4: Filtered and downmixed signal

2. Optics

2.1 Optic fp 50 shift Probe

There are different versions of the flow point Laser Doppler velocimeter, in order to cover a large range of applications. Different working distances are possible by using front lenses of different focal length. The front lens is mounted in a holder, which can be easily interchanged for other one with a different focal lengtth

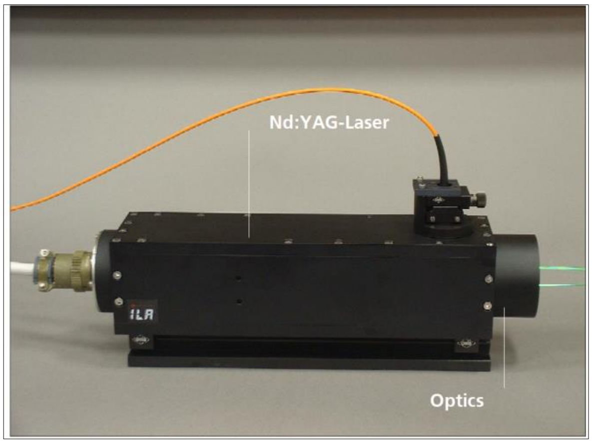


Figure 5: LDV probe fp 50 shifted

The LDV Probe fp50 shift is depicted in Figure 3 and Figure 5 with its main components.

The probe is easy to use and present high measurement accuracy. The Nd:YAG-laser head is directly integrated into the probe optics, without optical fibers. This results in two benefits: firstly, the laser light is transferred to the measuring volume without significant loss, thus almost all of the available laser power goes into the measuring volume. Secondly, there is no need for fiber couplers, which need to be adjusted periodically.

The probe is built with a path-length-compensated beam splitter, which generates two sending beams at a distance of 45 mm. Following this, one of the two beams is shifted by 40 MHz as it goes through a Bragg cell. The beams are adjusted by a set of wedge

prisms. Both beams are expanded by internal collimation optics, in order to reach a small waist size at the measurement volume. Finally, the beams are focused by the front lens into the measuring volume. The light scattered by the particles inside the measuring volume is collected backwards by the receiving lens and focused onto the end of the receiving fiber. All optical elements are rigidly fixed inside the probe. This and the robust mechanical and optical design of the probe guarantee high system stability and a high measuring accuracy.

2.2 Probe fp50 Adjustment

The LDV probe is mounted on an optical bench using two mounting brackets, which allow the user to rotate the probe around its optical axis. If the probe is removed from the optical bench or transported, the position of the receiving optical fiber might need to be adjusted again.

ILA GmbH
Karl-Heinz-Beckurts-Str. 13
D-52428 Jülich
Tel.: +49 2461 690-430
Fax: +49 2461 690-439
info@ila.de
www.ila.de

One side of the receiving fiber must be plugged into the SMA connector at the backside of the controller. The other side of the fiber is mounted on an X/Y translation stage at the top of the probe. By using this translation stage the end of the fiber can be positioned very precisely at the focus point of the backscattered light collected from the measuring volume.

If you use a probe with a calibration certificate of the PTB the xy translation stage is already adjusted and fixed by ILA: Do not demount or change the position of the xy-translation stage, because this could influence the calibration constant between 0,1 – 0,5%.

Step One: Preadjustment: To preadjust the receiving optics a diffuse reflecting surface, i.e. a business card, is positioned in the measuring volume, perpendicular to the optical axis of the probe. By disconnecting the fiber from the SMA-connector the reflected light from the measuring volume can be observed at the fiber end which is normally connected to the photomultiplier. By changing the position of the fiber with the translation stage adjustor screws, iterating from one to the other, the amount of light transmitted through the fiber can be maximized. This procedure is depicted in Figure 6.

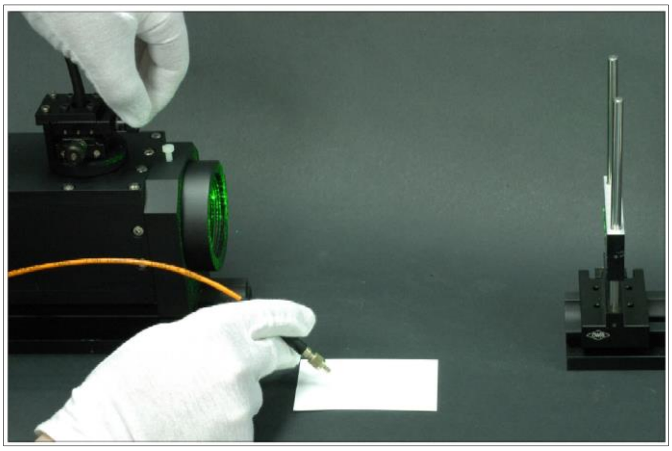


Figure 6: Preadjustment of the receiving optics

Step Two: Fine adjustment:

1. Connect the raw output of the signal to an oscilloscope or to the spectral analysis module.

2. Position the measuring volume inside the seeded flow field.

3. Optimize the amplitude of the modulated Doppler frequency by adjusting the X/Y translation stage.

NOTE: The raw signals are negative voltages. Please take this into account by choosing a negative trigger level.

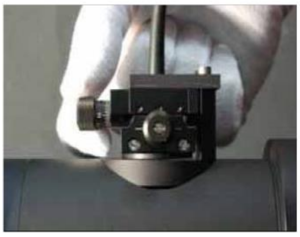
3. Laser

LDV systems need linear polarized single mode Lasers as light sources. The beams must have a high Gaussian quality. Gas lasers meet this requirement, but on the other hand they present a low efficiency and a large size.


Adjust the receiving optics.

LDV systems equipped with the fp50shift optics are usually equipped with Nd:YAG Lasers. They are available in a power range from 50 mW up to 500 mW, different wavelengths and mounted directly in the probe. The laser head power supply is housed in the LDV controller.

There is also the possibility to upgrade the system into a two components LDV system. For this purpose, an additional laser with a different wavelength is required (e. g. Nd:YAG laser).


Preadjust the receiving optics.

4. Controller

The main components of our LDV systems are the probe and the controller. The controller is based on a field-programmable gate array (fpga) and has been developed for unshifted, 1D and 2D LDV shifted systems.

It is equipped with a photomultiplier (PM) or an avalanche photo-diode (APD) depending on the customer's application.

The device is controlled through the LDA CONTROL QT software via Ethernet cable. It is possible to control the device remotely by connecting the LDV controller to a network, which is very useful in certain applications where the measurement area is restricted or located far from the controlling room. Figure 7 depicts the 1D LDV Controller.

Figure 7: 1D-LDV Controller

The controller can be configured to have analog high precision current/frequency inputs. This feature allows you to connect other devices to the controller, in order to consider additional physical conditions and/or measurements. For instance, temperature, velocity measurement with other devices such as Magnetic Flow Meter can be simultaneously acquired while performing the LDV measurement. Please refer to chapter 8.6.4 for further information.

5. Connecting the System

The LDV System consists of the following four main components:

- LDV Controller
- LDV Probe
- Computer and
- Traverse Controller.

The probe is connected to the controller through the main cable, which provides the power supply for all its components. The backscattered light is transferred to the photo detector in the controller using an optical fiber.

The LDV controller is connected to the computer via an Ethernet cable, which allows the communication with it locally or in a network.

The traverse controller is also connected to the computer via an Ethernet cable, and to the traverse units with D-sub cables.

We can divide the connection of the system in three main steps:

- connection of LDV controller and probe
- connection of LDV controller and computer and
- connection of the traversing system to the computer.

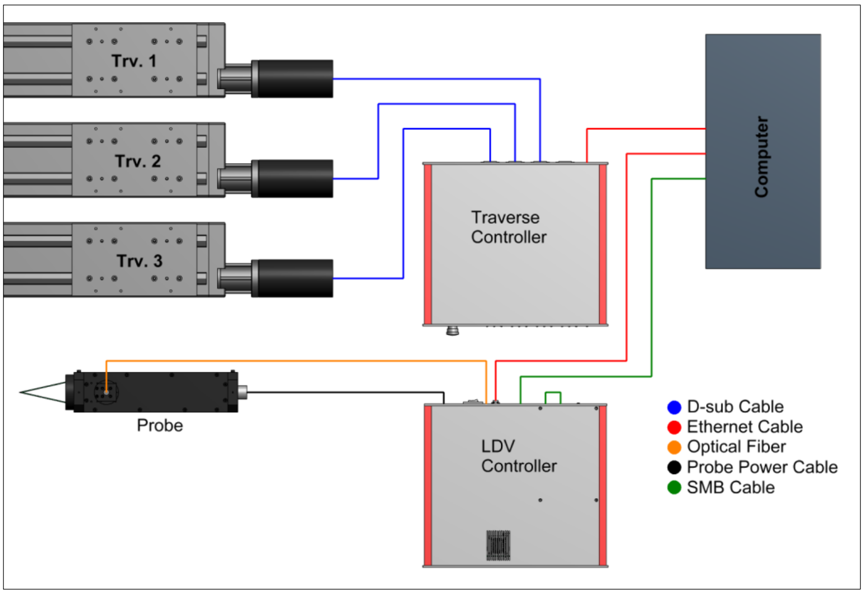


Figure 8: Connection diagram of the LDV system

5.1 Connecting LDV Controller to Computer

All the controller's connections are depicted in Figure 8. Please, follow the next steps in order to connect all the components (Figure 9):

- Connect the controller to the PC by using the crossover Ethernet cable
- Connect the "Controller Output", with the long SMB-SMB cable, to the AD-Card input at the rear panel of the computer (use the connection "CH0" for 1D LDV probe)
- Use the short SMB-SMB cable to connect the "Detector Output" to the "Controller Input" at the LDV Controller
- Connect the Controller to the power supply.

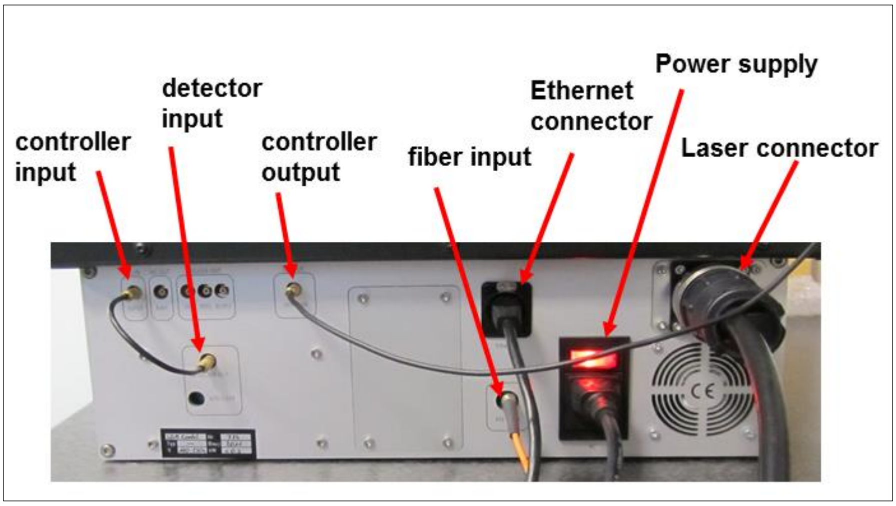


Figure 9: Back view of the LDV controller with all connected cables

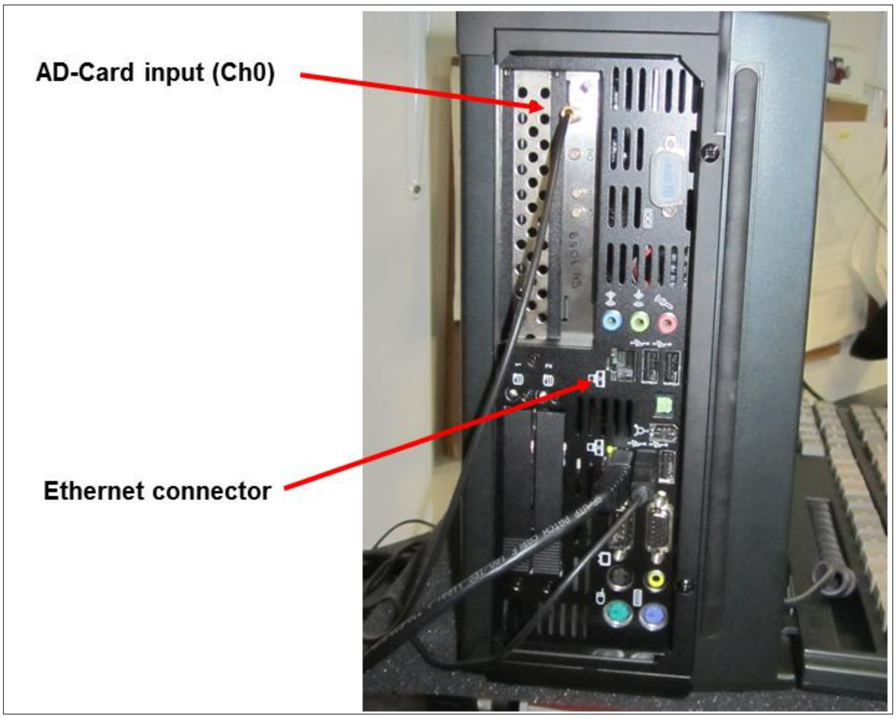


Figure 10: Back view of a computer tower with the connected cables

5.2 Connecting Laser Probe to LDV Controller

The Probe has to be connected to the LDV Controller, in order to power it on and to control all its parameters. Please follow the next steps:

- Connect the Probe to the LDV-Controller using the probe power cable and
- Connect the optical fiber end to the "fiber input" SMA connector at the backside of the LDV-Controller.

Once these steps have been performed, the LDV Probe is ready to be used. Figure 11 depicts the Probe, which is connected to the main cable and optical fiber.

Figure 11: Probe with xy-table, optical fiber and connected cable

5.3 Connecting Traversing System

The connections are depicted in Figure 9. Please, follow the following steps to connect all the components:

- Connect the linear units to the traverse controller using the D-sub cables
- Connect the Traversier Controller to the computer via Ethernet cable (second Ethernet slot) and
- Connect the controller to the power supply.

Figure 12: Traverse controller backside

ILA
Intelligent Laser Applications GmbH
ILA GmbH
Karl-Heinz-Beckurts-Str. 13
D-52428 Jülich
Tel.: +49 2461 690-430
Fax: +49 2461 690-439
info@ila.de
www.ila.de

6. Software Installation

Depending on the hardware present in your system, different burst-analysis-software modules LDA CONTROL QT are available. LDA CONTROL QT is running under Windows XP and higher Windows systems (e. g. Windows 7 and Windows 8). The Pentium PC workstation running LDA CONTROL QT should have at least 1024 MB of RAM. The A/D converter card (200 MHz or 50 MHz spectral analysis module) is mounted in one of the PCI Slots of the PC workstation.

Having inserted the installation CD, the LDA CONTROL QT Setup utility should start automatically. If the Autorun feature in the PC is not switched on, the software can alternatively be installed by running the following command:

CD-ROM-LW: \setup.exe

The software will automatically install in the C:\Programme\ILA\ subdirectory. Setup.exe will create its own icons in the main menu. The CD also contains the complete documentation and LDV training presentations can also be consulted directly on the LDA CONTROL QT CD, in the \Manuals directory.

ILA GmbH
Karl-Heinz-Beckurts-Str. 13
D-52428 Jülich
Tel.: +49 2461 690-430
Fax: +49 2461 690-439
info@ila.de
www.ila.de

7. Operating

To make measurements with LDA CONTROL QT, first connect the equipment as described in Chapter 5. With Windows running, the LDA CONTROL QT-software can be started by clicking on the LDA CONTROL QT - icon. The software's main function is to analyze the incoming Doppler burst signals using the spectral analysis module. The system can accept incoming Doppler burst signals from any LDA system.

To adjust signal parameters to your measurement, we recommend the following procedure.

7.1 General LDV Procedure

1. Open the LDA CONTROL QT software and produce and store a new *.ldv file
2. Get your interface right for your case (e. g. seeded flow)
3. Set up the different parameters in the menu "Option"
4. Set up the signal parameters in the "Property" window
5. Save the preselected LDV settings as *.ini file
6. Perform test measurements
7. Adjust parameters in steps 3 and 4 until results satisfactory.

7.2 Step by Step

Before starting measurements, make sure that the measuring volume is positioned in the flow field and that there are seeding particles in the flow.

Once LDA CONTROL QT is started, measured velocity data can be stored in an *.ldv file (**Figure 13**). The *.ldv file contains all the positions of the measurement points and the measuring values (average velocity and turbulence degree) associated with these positions.

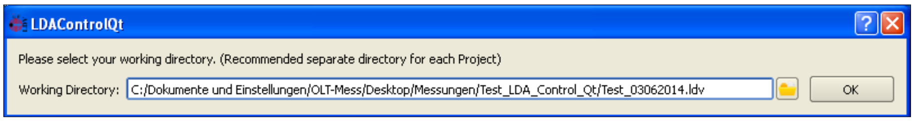


Figure 13: Screen for selecting the working directory and the name of the *.ldv file

The Doppler frequencies of all Doppler bursts, the arrival times of all bursts and the most important parameters (e. g. interference fringe distances and shift frequencies) are stored in a separate *.ch0 file (and *.ch1 file if two velocity channels are used). The same information is also stored in a file with the csv format. The *.ldv file can also be created with the "Save Mess" command in the course of operating software.

After selecting the working directory or quit these menu with "OK" you should see a screen like shown in Figure 14.

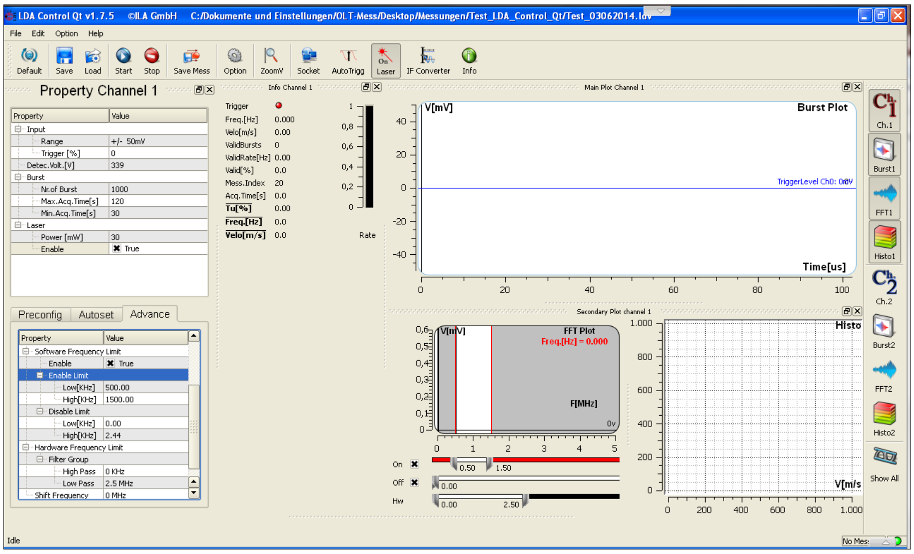


Figure 14: Start screen of the LDV software

You will see different frames (from left to right):

- the "Property Channel 1" window with two tables with a lot of parameters for the measurement

- the "Info Channel 1" window with a summary of the most important results of a measurement
- the "Burst plot" window, showing the transient Doppler signal
- the "FFT plot" window with the calculated frequency spectrum
- the "Histogramm" window with the actual distribution of the measured frequencies or the measured velocities and
- a lot of small buttons for the different windows on the right side of the screen.

Clicking the button "Option"  will open a new window with a lot of submenus (Figure 15).

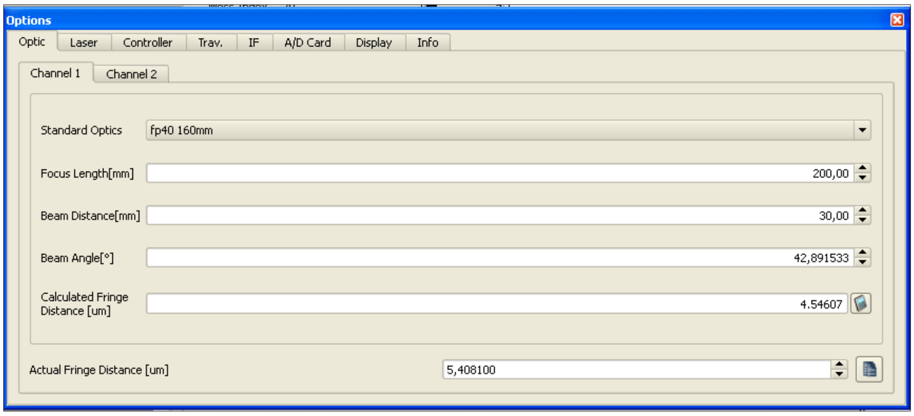


Figure 15: The Option Menu with the Optic submenu

Please choose at first the submenu "Laser" in order to select the used laser source of your system or to give in the wavelength by hand (Figure 16).

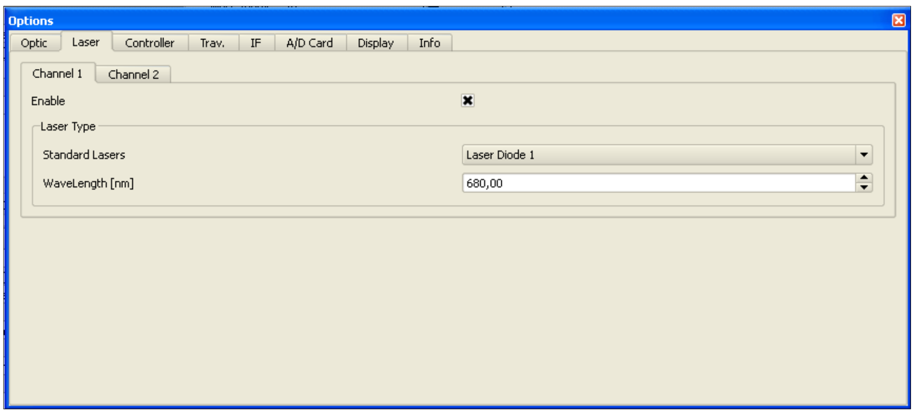


Figure 16: The Option Menu with the Laser submenu

Now change back to the submenu “Optic”. Here you have to set the calibration factor (interference fringe distance) of your LDV probe. It is possible to calculate the calibration factor with the help of the theoretical values of the wavelength of the used laser source and the parameters of the transmitting optic of the LDV probe. If you know the exact value of the interference fringe distance from a special calibration procedure you should overwrite the value in the field “Actual Fringe Distance”.

Now you should select the submenu “Controller”, where you can check the IP address of your controller and enable the Bragg cell and the downmix frequency if you use a shifted LDV system with detection of the sense of the direction of the measured velocities. With the help of the signal gain factor you can amplify the raw signals.

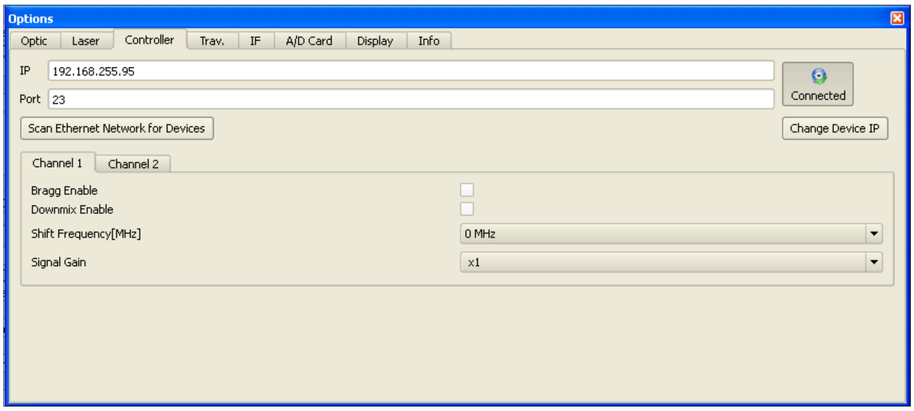


Figure 17: The Option Menu with the Controller submenu

To set up the shift frequency and also the following selecting of the band-pass filter the expected flow velocity needs to be estimated. In an example we expect a velocity of 7 m/s. With the fringe distance of Δx = 2.9675 µm the expected Doppler frequency $\mathbf{F_d}$ is calculated by

$$\mathbf{F_d = v / \Delta x}$$

$$\mathbf{F_d = 7\ m/s / (2.9675 \cdot 10^{-6}\ m)}$$

$$\mathbf{F_d = 2.35\ MHz.}$$

Intelligent Laser Applications GmbH
ILA GmbH
Karl-Heinz-Beckurts-Str. 13
D-52428 Jülich
Tel.: +49 2461 690-430
Fax: +49 2461 690-439
info@ila.de
www.ila.de

For shifted systems the expected signal frequency $\mathbf{f_{sig}}$ is the sum of the Doppler frequency and of the shift frequency $\mathbf{f_s}$ if a particle is crossing the measuring volume in the direction opposite to the direction of motion of the interference fringes:

$$\mathbf{f_{sig.1} = f_s + (v / \Delta x)}$$

In the opposite a particle moving with the motion of fringes yields a lower signal frequency $\mathbf{f_{sig}}$.

$$\mathbf{f_{sig.2} = f_s - (v / \Delta x)}$$

That means that you have to choose the shift frequency so that the shift frequency is higher than the expected maximum value of the Doppler frequency if you would like to measure velocities in the sense of direction of the moving fringe pattern.

For the first case a shift frequency of 5 MHz yields to the expected signal frequency of

$$\mathbf{f_{sig.1} = 5\ MHz + 2.35\ MHz = 7.35\ MHz}$$

Corresponding you have to set the downmixing frequency to 5 MHz for this example.

With the same downmixing frequency you can could also measure velocities in the other direction of sense because of

$$\mathbf{f_{sig.2} = 5\ MHz - 2.35\ MHz = 2.65\ MHz}$$

Now close the Option menu and go to the “Property Channel 1” window (see Figure 18).

Start with the settings in the upper table. For the input range begin with a trigger range of 50 mV and a trigger threshold of e. g. 3.125 %. Later you have to optimize these two parameters in detail.

You should set the detector voltage, for a photomultiplier (PM) to a value in the 900 – 1400 V range. Start with a value of 1000 V. For an avalanche diode the range is approximate between 340 V and 380 V.

With the self-explaining parameters of the next group “Burst” you control the duration of a measurement.

Some kinds of lasers (e.g. Nd:YAG lasers) allows a variation of the laser power from a minimum to a maximum. In the group “Laser” you can set this value and control the turning-on of the laser.

Safety Instructions
Measuring Principle
Optics
Laser
Controller
Connecting the System
Software Installation
Operating
Detailed Software Description
Appendix

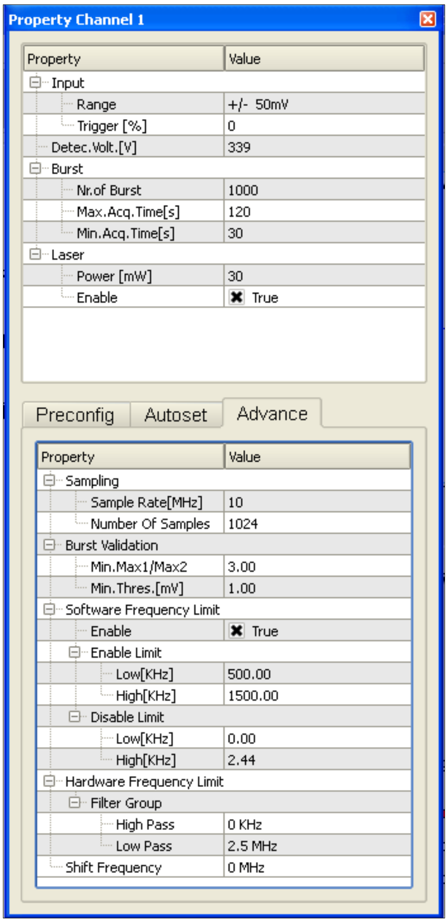


Figure 18: The “Property Channel 1” window

Now go to the submenu “Advance” in the lower part of the “Property” channel. There you have to choose the Sample Rate for the FFT. This value should at least ten times higher than the expected signal frequency, that means for an shifted LDV system approximately the Sample Rate should be ten times higher than the shift frequency (in our example e. g. appr. 75 MHz). A good start value for the number of samples is 1024.

In the group “Burst Validation” set the values for “Min. Max1/Max2” on 3.00 and for “Min. Threshold” on 1 mV.

For the first measuring tests disable the “Software frequency Limit”. Later you can use the software filters in order to optimize your signals

In the group of “Hardware frequency Limit” choose suitable values for High Pass and Low pass according the expected Doppler frequency. For our example set the high-pass filter to 3 MHz (lower than the expected Doppler frequency f_d) and set the low-pass filter to 10 MHz (higher than the expected Doppler frequency f_d). Check also the value of the downmixing frequency, which you have set in one of the preceding steps.

All the parameters you should store in a file with the extension *.ini. Therefore use the button “Save” . Now you could start the measurement pushing the button “Start” .

Once the measurement is started the transient input signal will be shown in the Burst Display window. If the A/D converter card does not trigger, one or several of the following parameters can be adjusted:

- trigger input range
- trigger threshold
- detector voltage
- high- and low-pass filter

Now you can see at the graphical windows some information. At the “Burst plot” window you can see the transient signal. On the histogram there are the valid bursts listed as a histogram. The FFT and the frequency histogram are shown at the “FFT” window. Using the frequency limits of the software filters (shown with blue lines) unwanted peaks found in the displayed frequency range can be excluded.

In the “Info channel 1” window you can observe the current average value of the Doppler frequency as well as the measured flow velocity and the number of valid bursts already processed.

1. If the bursts appear “cropped” in the Burst Display window this means you are only sampling part of the bursts: change the sample rate and/or the number of samples until the display shows clearly complete bursts.

2. If the system is triggering but you get no valid bursts you are may be triggering on signal noise: change to a higher trigger level.

3. If the system is triggering but you get no valid bursts your flow velocity might appear different from what you expected. To correct this, adjust the frequency limits in the lower table of the “Property” window in the software. Also try other sampling rates.

4. You can save the measured velocity values and the measuring positions in data file *.ldv with “Save Measure” . For every stored velocity value the histogram data will be saved in a separate file with the extension(s) *.ch0 (and/or *.ch1 if you use a 2D LDV system) and also in the .csv format.

Safety Instructions
Measuring Principle
Optics
Laser
Controller
Connecting the System
Software Installation
Operating
Detailed Software Description
Appendix

8. Detailed Software Description

8.1 The main Frame

The LDA CONTROL QT software is a Multiple Document Application, i.e. the application can spawn multiple windows showing different data. The main window (Figure 19) displays the transient signal in the upper part of the window, with the spectral analysis of the Doppler burst directly below.

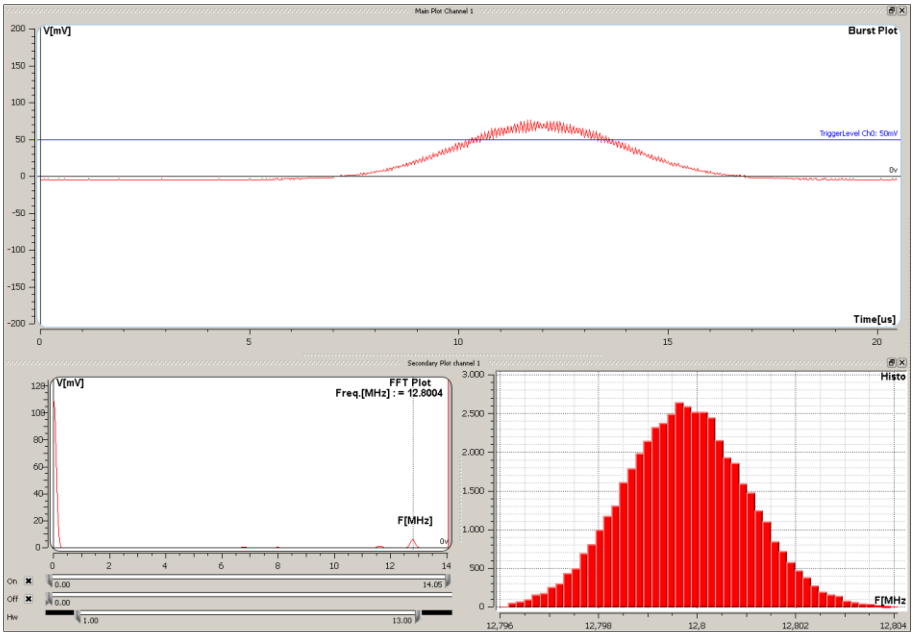


Figure 19: Unfiltered LDV-Signal

The frequency peak in the spectrum corresponds to the frequency of the burst that is currently being processed. The vertical cursors show the frequency limits - areas in which the Doppler frequency should be accepted (dashed lines), and areas which should be excluded (dash-dot lines). The position of the frequency limits can be changed in the signal processing window. Figure above shows a signal typical of an unfiltered (raw) burst, such as would be available at the RAW exit socket of the LDV controller.

The pedestal is clearly visible at the lower frequency end of the spectral analysis, while the modulated Doppler frequency (much higher at 12.8 MHz) can be seen higher up. Note that this window must remain opened – you can of course minimize it, but closing it will terminate the application.

8.1.1 Main GUI

Safety Instructions
Measuring Principle
Optics
Laser
Controller
Connecting the System
Software Installation
Operating
Detailed Software Description
Appendix

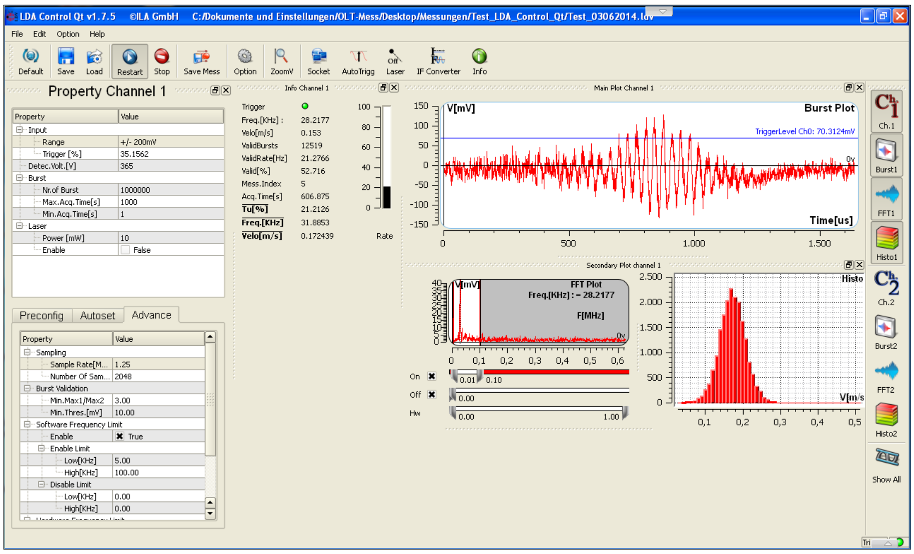


Figure 20: Main User Interface

Figure 20 shows the main User Interface (UI) of the LDA CONTROL QT-Software. This could be a typical view of the user interface during a measurement. In this case only one channel is acquired (channel1). The three important displays (Burst-Plot, FFT-Plot and Histogram) will be explained in detail in chapter 8.2, 8.3 and 8.4.

The control icons are set on the top and buttons for a variety of plot-windows are on the right. Clicking any of these buttons generates a separate window that can be placed and fit in the middle of the UI by drag and drop. That allows a free arrangement of the individual plots for your view.

The buttons on the top belongs to the Control Panel, which will be described in more detail in the following pages.

8.1.2 Control Panel

	Default: Clicking this button will makes a reset to default setting

	Save: In order to save the configuration of a measure-ment-job click the save button. This generates an *.ini-file which can be loaded for a further measurement with the same configuration.

The *.ini-File contains data as following:

1. all settings regarding the measurement made in signal processing window (see chapter 8.5)
2. the set-up of plot windows (see chapter 8.1.3)
3. the axis properties of all plot windows and
4. all properties made in the options menu (see chapter 8.6).

The *.ini-File does not contain any measurement data. The measure-ment data is stored as an *.ldv-file. After every measurement-job the user is asked where to save the *.ldv file. Data which is saved as an *.ldv-file has the appearance shown in chapter 8.9.1

	LOAD: To load en existing *.ini-File click this Button

	START: To start the measurement click this button. Make sure that the Laser is ON and pointing on the measurement volume. During the measurement this button appears grey indicating that the measurement is active.

	STOP: This Button quits the measurement immediately.

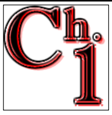
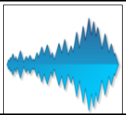
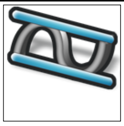
Safety Instructions
Measuring Principle
Optics
Laser
Controller
Connecting the System
Software Installation
Operating
Detailed Software Description
Appendix

	SAVE MESS: Saving the measurement as *.ldv-file
	OPTION: This button calls the Option-Panel. All Settings which can be made here are described in chapter 8.6 in more detail.
	ZOOM: Use this button to zoom in and out of plots
	SOCKET: This button can be useful for troubleshooting regarding network connection. It shows the commands that sent to the controller and can contain also the error messages.
	AUTOTRIGG: Pressing this button will set the trigger-level automatically. This might be helpful for initial capturing of bursts.
Off On	ON LASER: This button enables/ and disables the laser. This button gives also a status indication about the current situation. (Laser ON/ OFF). Before starting the measurement make sure the laser is on.
	IF CONVERTER: This button calls the option to acquire analog signals. This option allows to sample other analog inputs such as (temp./ pressure/flowrate-measurement) along with LDV-Measurement in one *.ldv-file. For detailed description of the function about the Settings go to chapter 8.6.4
	INFO: Click here to see information about the software version and other information about the program.

Intelligent Laser Applications GmbH
ILA GmbH
Karl-Heinz-Beckurts-Str. 13
D-52428 Jülich
Tel.: +49 2461 690-430
Fax: +49 2461 690-439
info@ila.de
www.ila.de

8.1.3 Plot Control

The Plot-control buttons are placed on the right side. This group of buttons contains the channel button, the Burst Plot, the FFT Plot and the Histogram. When the channel-button is clicked the corresponding Burst-plot, FFT-plot and Histogram will pop up.

	All three windows for channel 1 will pop up
	Burst window
	FFT window
	Histogram window
	Display-All

8.2 Burst window

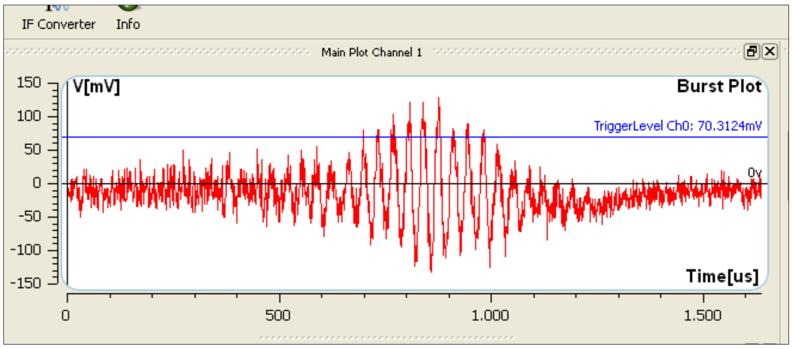


Figure 21: Burst Window

The Burst window shows the valid bursts during a measurement. The main function of this plot is to capture the burst along with all settings explained in signal processing window in chapter 8.5. This plot contains the signal of a burst that will occur when the measurement volume is passed by a particle. It is predefined to center the valid burst. The signal tracked in this plot is the filtered and amplified voltage of the photo detector, basically it is voltage over time .When a burst is triggered the software centers the burst in the chosen time frame (see for more details chapter 8.5). The velocity information is acquired from the backscattered light. To increase the signal amplitudes photomultiplier (PM) voltage can be increased. The PM voltage starts at 50 and stops around 2200 Volt, depending on the photomultiplier type. Changing the PM voltage amounts the control of the gain of the PM. The higher the voltage the higher the gain will be. With a high gain the noise will also increase. When the amount of backscattered light exceeds a certain limit the PM turns into saturation mode, so the burst detection will not be possible. If too much backscattered light is reaching the PM the light can be limited with an aperture which can be mounted in front of the lens. As described y-axis represents the voltage and x-axis the time. In the display-options-menu that is described in chapter 8.6.6 the axis can be scaled. To optimize the valid burst signals there are two important parameters (range and trigger level).

ILA GmbH
Karl-Heinz-Beckurts-Str. 13
D-52428 Jülich
Tel.: +49 2461 690-430
Fax: +49 2461 690-439
info@ila.de
www.ila.de

RANGE: This value defines the input voltage-range of the A/D converter. The resolution of the burst is dependent on the chosen range. For example, if the A/D converter has a resolution of 8 bit and the range is set to +/- 1 V ((+/-1 V => range :2V)) the quantisation of the analog signal would be 2V/ 2^8 = 0.0078125 V. If there is a large mismatch between range and input the signal will not be quantified in a reasonable manner. The input signal voltage is highly interrelated with many conditions as laser power, PM-voltage, intensity of backscattered light, focal distance and furthers more. Hence, the range should be adapted individually to the measurement condition.

TRIGGERLEVEL: In order to capture clean burst signals which will be accounted for the measurement, the trigger level should be set mindfully. The trigger level is a voltage threshold that should be exceeded by the signal to validate a Burst. In that way only signals will be taken into account that exceeds a certain noise-level. The trigger level can be chosen interactively in the plot by dragging the blue line in the Burst plot or defined as a decimal value in property channel (See chapter 8.5).

In order to capture bursts it is important to find the correct sample-rate and timeframe. The duration of a burst is strongly coupled with the measured velocity. Logically high velocity leads to short Burst-duration and low velocities to long duration. The Sample Rate should be set 5 to10 x larger than the expected Doppler frequency. The timeframe results from sample rate and number of samples. The sample Rate und number of samples should be chosen in that way that the burst clearly fit in the timeframe. The number of samples goes directly into the FFT calculation. If this value is chosen to high with a relatively small burst and a large amount of noise will be captured, the FFT calculation will not be processed effectively. The large amount of noise which will be processed in FFT is useless and time-consuming as well. By setting number of samples to capture the burst with less amount of noise will make the FFT processing much quicker so that more bursts can be processed in less time.

8.3 FFT Window

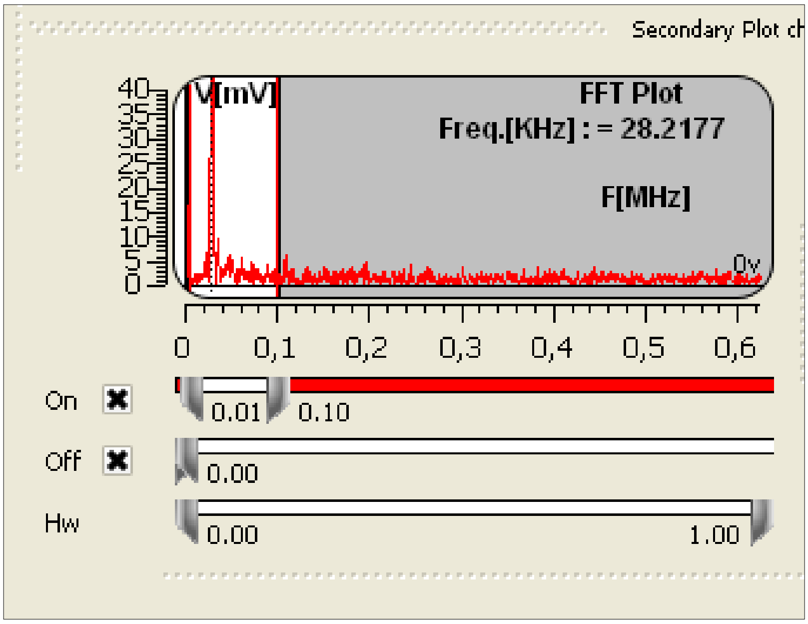


Figure 22: FFT Window

The validated bursts are displayed continuously in the burst plot. To acquire the frequency of the burst which contains the Doppler frequency a FFT is processed. The FFT transforms the time valid signal in frequency spectrum. The window size of the FFT is equal to the number of samples as described before in chapter 8.2. The FFT plot reveals the frequencies with the corresponding amplitudes of the signal. By making an assumption of zero-noise the FFT should only reveal one peak at the Doppler frequency. In reality noise is present in all frequencies as depicted in

To isolate the Doppler Frequency in FFT-plot software filters are built in. That allows to cut off unwanted disturbance frequencies

Figure **22**).

Figure **22** shows two bars with „ON“ -> Inclusion and „OFF“-> exclusion of selected bandwidth of frequencies. Excluded frequencies will not be included in the measurement even if the amplitude reaches a high value. Here is also the possibility given to change the bandwidth of hardware filters. (See more about HW-filters in chapter 8.5).

One important criterion to validate bursts is the ratio of amplitudes within a FFT-Window.: Experience revealed that an amplitude ratio min. Max1/Max2 = 3 is a good value to assure that only valid frequencies, which is dominating the FFT-plot is taken into account. Amplitudes which are not 3 times higher than the other amplitudes will not be considered. This set up is made in signal processing window (see for more details chapter 8.5).

The FFT procedure gives a choice of standard windowing functions: the options are Blackwell, Hanning, Hamming and no windows (square). These functions are used to minimize the leakage effect of the FFT. The Hanning method is implemented in the software. The peak fitting interpolation in the FFT is based on a Gauss interpolation, so that this interpolation is implemented hence it brings the best results (see for more details chapter 8.5).

8.4 The Histogram Window

The histogram window shows the number of valid bursts as a function of the velocity. It is the sum of all detected valid Doppler frequencies. The slowest particles are shown on the left and the fastest on the right side of the graph.

Figure **23** depicts the valid burst in terms of Doppler frequency, alternatively velocities can be shown instead of frequencies. (See chapter 8.6.6 how to change to velocities). By measuring velocities with high accuracy, the histogram will reveal a Gaussian distribution with a low mean variation as shown in

Figure **23**. The frequencies are classed automatically, if needed the class length can be set manually by selecting the number of columns (see chapter 8.6.6).

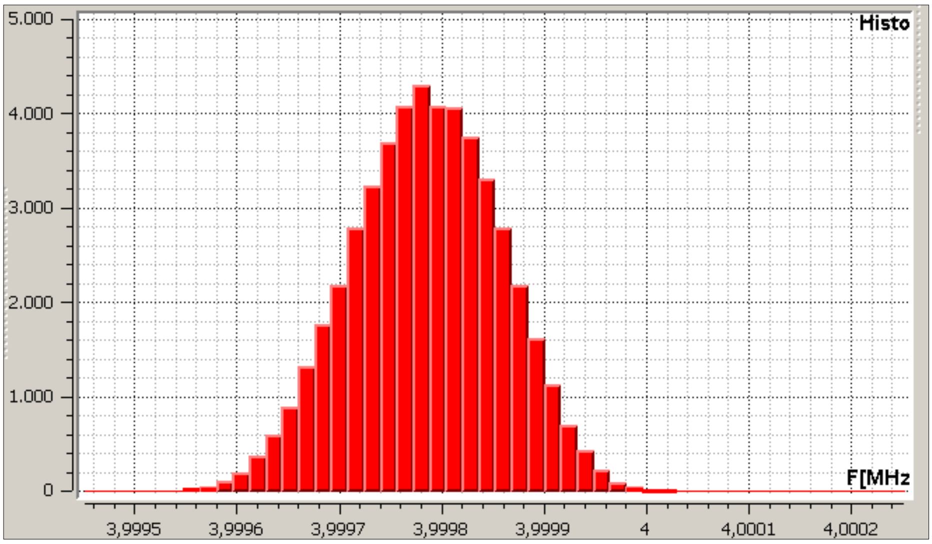


Figure 23*:* **Histogram Window**

While measurement is taking place the Histogram can be reset by clicking on the restart-button which will be visible when the measurement is active.

ILA GmbH
Karl-Heinz-Beckurts-Str. 13
D-52428 Jülich
Tel.: +49 2461 690-430
Fax: +49 2461 690-439
info@ila.de
www.ila.de

The LDV Data Frame

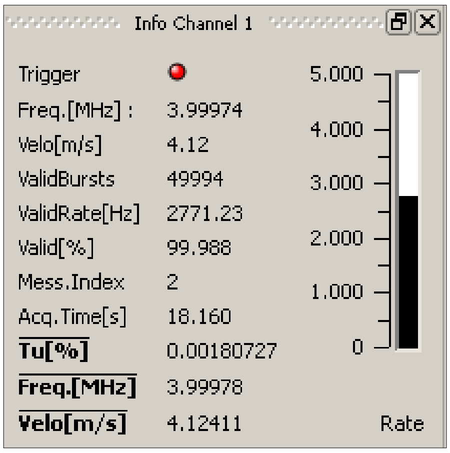


Figure 24: LDV Data Frame

The actual velocity value, the Doppler frequency and the degree of turbulence are shown in this window (Figure 24). Furthermore it gives information concerning the trigger condition, data rate, data validity, acquisition time, etc. If the trigger status is green, a trigger event is taking place. This status light should always flicker while measuring.

ILA GmbH
Karl-Heinz-Beckurts-Str. 13
D-52428 Jülich
Tel.: +49 2461 690-430
Fax: +49 2461 690-439
info@ila.de
www.ila.de

8.5 Signal Processing Window

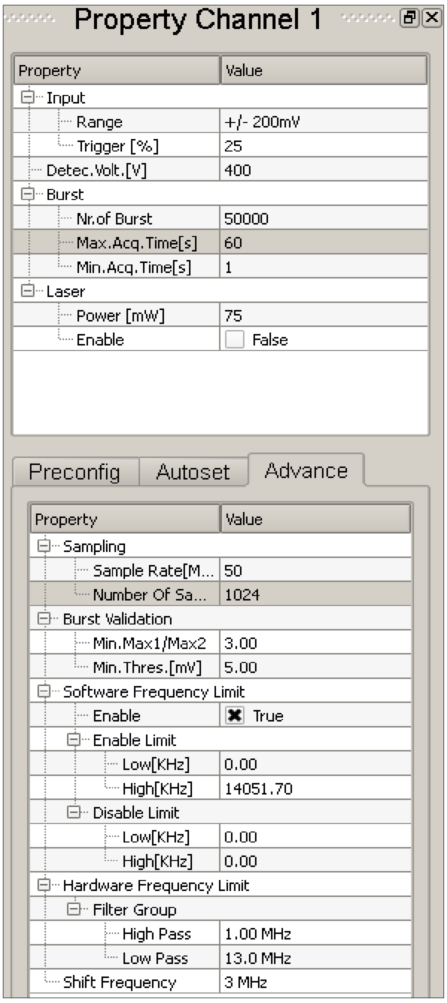


Figure 25: Signal processing window

Safety Instructions
Measuring Principle
Optics
Laser
Controller
Connecting the System
Software Installation
Operating
Detailed Software Description
Appendix

The main set up for signal processing is made in this window. As described in chapter 8.2 range, trigger level and detector voltage can be set in this menu. The setting-block (Burst) defines the limit of one measurement. The measurement will quit when the number of bursts **AND** Min.Acq.Time criterions are fulfilled **OR** when the max.Acq.Time has been reached. In the next block, the laser power can be set and enabled. Make sure that the laser is enabled before starting the measurement. In the advanced settings important parameters for capturing and validating signals are done. The first parameter is the sampling frequency.

The ("Sample Rate"), which should be set, as described before, at 5 to 10 times higher than the expected Doppler Frequency. The sampling frequency is always the same for both channels. On some A/D converter cards (flow200 und flow400) two channels can be merged using an interlacing technique to achieve high sampling frequencies, up to 200 and 400 MHz (interlacing technique) on the resultant single channel. The samples parameter is the number of time samples selected for both channels. The FFT is calculated using the sampled time series of the Doppler signal. Following can be seen: The higher the number of samples, the higher the accuracy of the FFT, but the lower the overall data rate, because of the higher processing time. With an AMD processor (1.4 GHz) and 256 selected as the number of samples, data rates up to 20 kHz are possible. For 1024 samples, this figure goes down to about 1 kHz, depending on the signal quality and the seeding conditions. The accuracy of the FFT is influenced by the combination of the sampling frequency and the number of samples. To achieve a high accuracy, the FFT needs a high resolution FFT, so a minimum of 512 samples should be used. The sampling frequency should be selected so that the complete burst fits in the display window. The number of samples is a multiple of 2. The maximum amount of samples is 16 MB.

In the next block, burst validation criteria in frequency domain are set. As described in chapter 8.3, the experience revealed to have good results to set min.Max1/Max2 value to 3. The next condition sets the Min.Thres. to assure for only taking frequencies out of the noise.

The next block contains settings of the filters. Basically two filter types are in use: software- and hardware-filters. The software filters are used for the frequency domain and hardware filters for the time domain.

The hardware filter is a bench of analog filters which are placed in the LDV- controller. With a free set up of the high- and low passes (HP,

ILA GmbH
Karl-Heinz-Beckurts-Str. 13
D-52428 Jülich
Tel.: +49 2461 690-430
Fax: +49 2461 690-439
info@ila.de
www.ila.de

LP) it is possible to have a wide range and variety of band pass arrangement. The selection of the filters (HP, LP) can be done quickly by knowing roughly the velocities which should be measured. For instance when the velocity is about v=1 m/s and the calibrated fringe distance is given as d = 1 µm the expected Doppler frequency will be f = v / d = 1 m/s / 1µm => 1 MHz. Ideally the band pass should be centered at 1 MHz for this case. By having a shifted system the Doppler frequency will be added or subtracted to the shift.-Frequency depending on the sense of flow direction (see also chapter 7.2).

It is logic to set the hardware filters as first and no other way around. The software filters has also the feature to exclude a frequency range, so that some unwanted signals can be excluded in the frequency domain.

All these settings can be stored as an INI-file and can be loaded for an upcoming measurement. Along with these setting the Doppler Frequency is processed as following.

1. The input signal exceeds the trigger level.
2. If the maximum time for the measurement is not exceeded, FFT processing of the signal is started.
3. Calculation of the frequency spectrum.
4. Calculation of the amplitude ratio, i.e. the proportion of the highest and the second highest amplitude in the frequency domain.
5. The burst is considered valid if the following conditions are fulfilled simultaneously:
    - the amplitude ratio is higher than min Max1/Max2 and
    - the modulated amplitude of the burst is higher than the threshold set in the Signal processing window and
    - the Doppler Frequency is inside the valid frequency and limits.

8.6 OPTION

Options that belong to system parameter or visualization parameter are summarized in the Options Menu. The detailed description of the adjustments that made here will be presented as following.

8.6.1 Optics

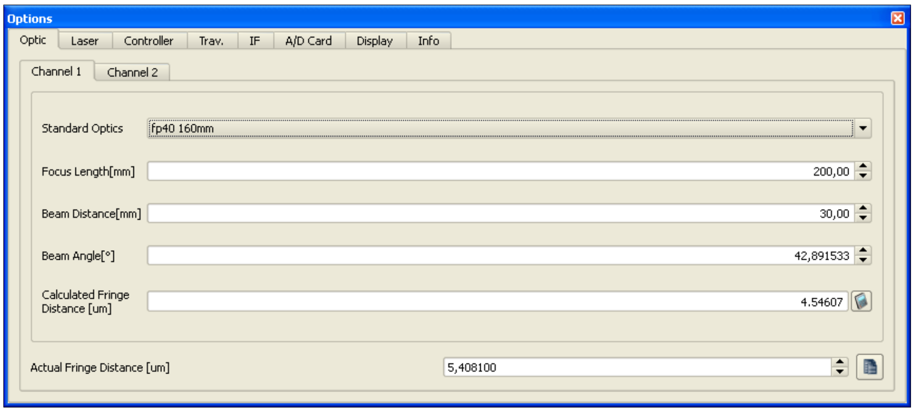


Figure 26: Option-Optics

It is very important to set the correct calibrated fringe distance in this frame. Otherwise the velocities might be wrong. By setting the focus length, beam Distance and beam angle the software calculates the theoretical fringe distance. To be more accurate it is recommended to use the calibrated fringe distance which will be commonly acquired from the manufacturer.

The corresponding values for a variety of probes and lenses are already predefined and can be selected from the drop down bar: Standard Optics.

8.6.2 Laser

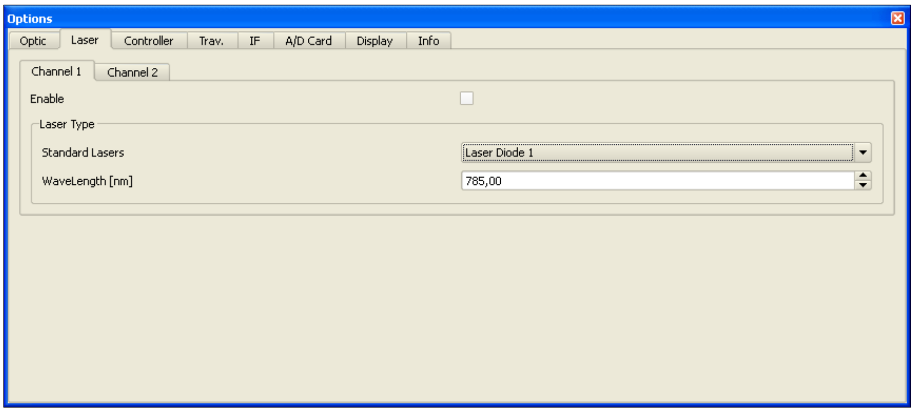


Figure 27: Option-Laser

In this tab the laser that is used for each channel is defined. Specify the laser type and the wavelength correctly. This information can be found normally on the LDV probe.

8.6.3 Controller

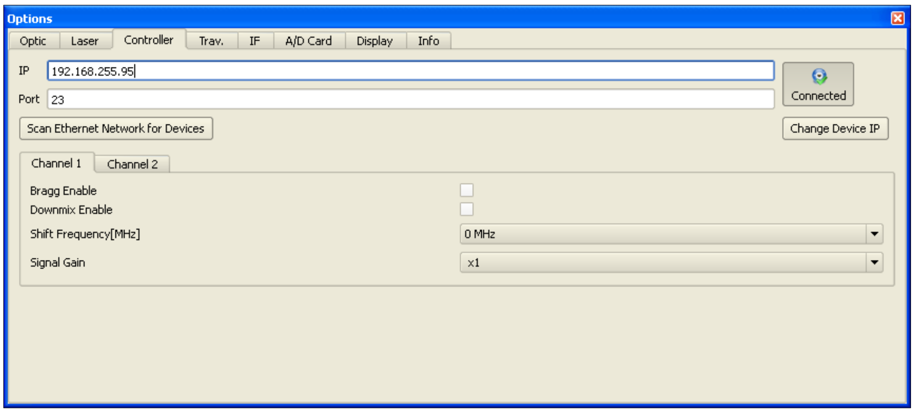


Figure 28: Option-Controller

The IP-address setting of the controller has to be done in this frame. The IP-address of the controller can be allocated manually. The button besides the bar (Connected /Disconnected) allows the user to connect/disconnect the controller with the allocated IP address. There is also the feature to scan for devices in the local network.Shifted systems must enable Bragg cell and downmix and the shift frequency should be set.

8.6.4 IF

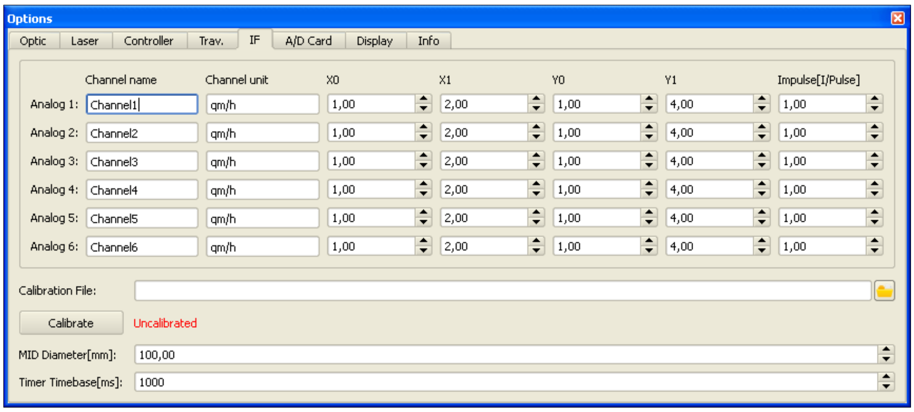


Figure 29: Option-IF

This tab is to configure the analog inputs of the controller. It contains the declaration of input-channels and scaling. At first define the channel name and unit. For example channel 1 is connected to the output of a flowmeter with the dimension [qm/h], then enter the signal range in X0 and X1, for instance X0 = 4 mA, X1 = 20mA. Enter the corresponding physical measurement values for example Y0 = 0 qm/h and Y1 = 100 qm/h. The analog inputs will be scaled with this factor and recorded along the ldv measurement in the corresponding channel column in the LDV file. Some devices deliver pulse output. Here you can define the scaling factor for pulses for example in terms of flow measurement 10 liters per pulse. Each controller is equipped with an specific A/D card for the analog signal tracking. Therefore the correct calibration file for the particular A/D card should be loaded and calibrated with the associated Button: Calibrate.

8.6.5 A/D Card

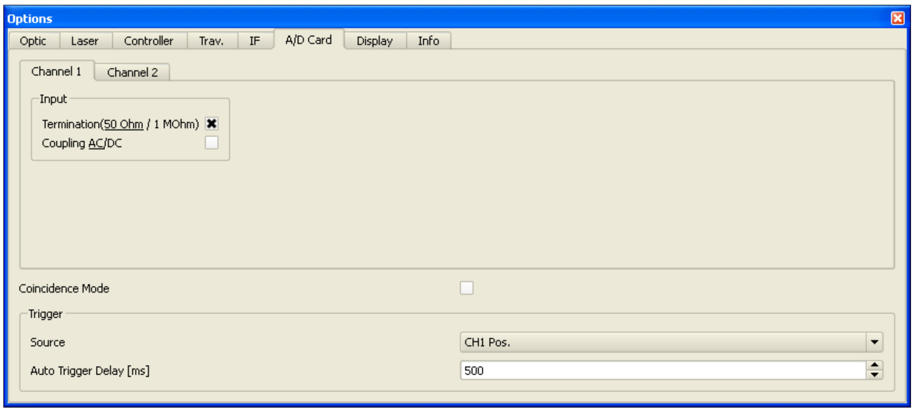


Figure 30: Option-A/D Card

Settings for the A/D contains the termination and coupling, which will be implemented in a future version. The checkbox named coincidence mode is important for validation of a measurement point (MP) for two channel measurement. If coincidence mode is enabled only incoming valid bursts of both channels will be taken. If channel one triggers a non-valid burst and channel two triggers a valid burst no velocities will be recorded. When coincidence mode is disabled, whenever a valid burst receives also the velocity-reading of the other channel will be taken for the measurement. Trigger source allocates the channel for triggering. Auto trigger delay defines a time period that is needed to set the trigger level automatically.

8.6.6 Display

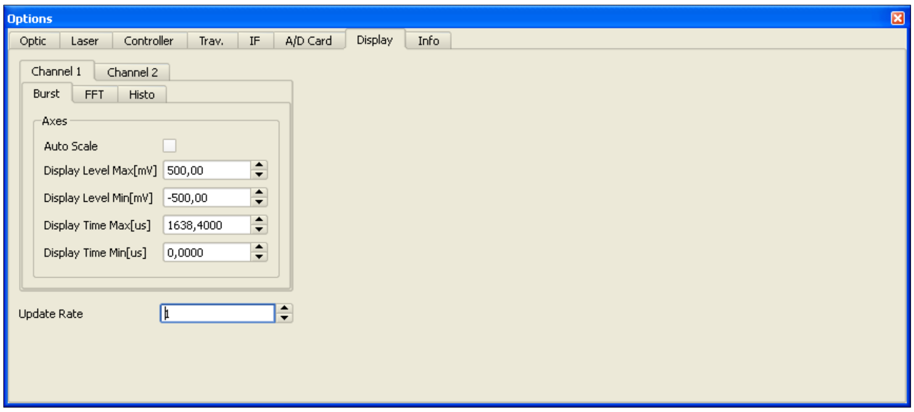


Figure 31 Options- Display-Burst

Settings to visualization of the different plots are made here. This Tab will also pop up by right click on any plot to set the scaling of the axis.

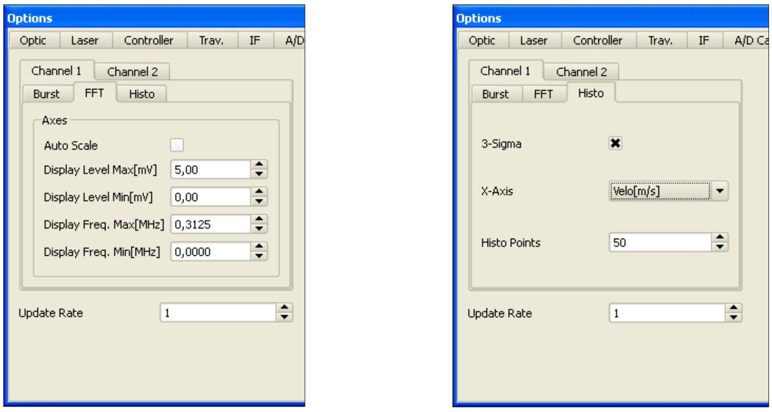


Figure 32: Options- Display-FFT & Histo

Update rate defines how often the plot is updated. If 3 is selected the plot will be updated after every 3rd burst.

In Histo tab there is a checkbox called 3-Sigma, enabling this checkbox will cut off velocities which lie beyond three sigma away from the mean value before storing the measurement. The columns of the histogram can be selected depending on mean variation of the velocity.

Intelligent Laser Applications GmbH
ILA GmbH
Karl-Heinz-Beckurts-Str. 13
D-52428 Jülich
Tel.: +49 2461 690-430
Fax: +49 2461 690-439
info@ila.de
www.ila.de

8.7 Traverse Menu

In a wide range of LDV-Application it is desired to measure a velocity flow field. Therefore the area of measurement has to be discretised with measurement points (MP). The possibility to move to each MP is commonly conducted with a traversing unit.

Clicking the button 

Traverse on the control panel leads to the traversing menu where you have a bunch of possibilities to put a measurement grid and to move the probe to each individual MP.

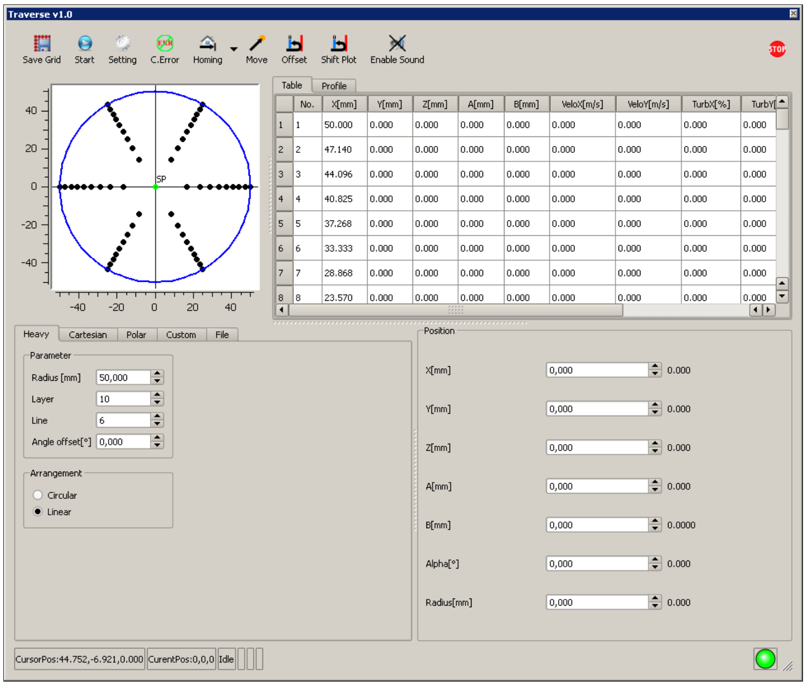


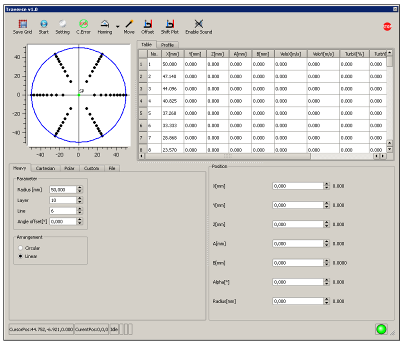
Figure 33: Traverse Menu

The usual procedure to perform an real measurement is to choose a grid and make the measurement along the grid. The grid can be set manually or loaded from a file. The LDA CONTROL QT software can generate different kind of measurement grids such as heavy Grid, cartesian and polar grid. In following the grids will be described in detail.

8.7.1 Heavy Grid

For volume flow metering, the use of heavy lines is very common. In this case, the measurement points are positioned on the gravity lines of concentric surfaces with constant areas. So the density of the grid is higher at the periphery. Heavy lines accept a radius, a number of layers and a number of lines. If the parameters are changed, the positions of the individual measurement points will be recalculated. There is also the possibility to apply an offset angle to the measurement grid; in this case all the point coordinates will be recalculated.

Figure 34 shows a heavy grid with a radius of 50 mm, 10 layers and 6 lines. Simultaneously, on the right side it is shown the same grid rotated by an angle of 10°.



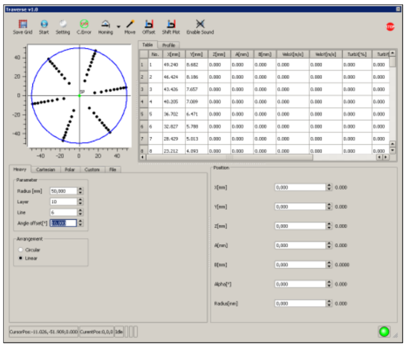


Figure 34: Heavy Grid

8.7.2 Cartesian Grid

In many cases it is useful to work with a cartesian grid. The parameters for the setting of cartesian grids are the width, height and the distances between the MPs in each direction (Hor.distance and Vert.distance). The green point in the plot is the origin of the grid. The orange point indicates the selected point (SP). Each point can be selected individually by clicking on it. The selected point is highlighted on the table with the corresponding MP-Nr. To move the probe to this particular point click on the move button. That will lead the probe traverse to the selected point. If you want to change the position of a particular MP, go to the plot and click on the appropriate point and change the x and y to the desired value.

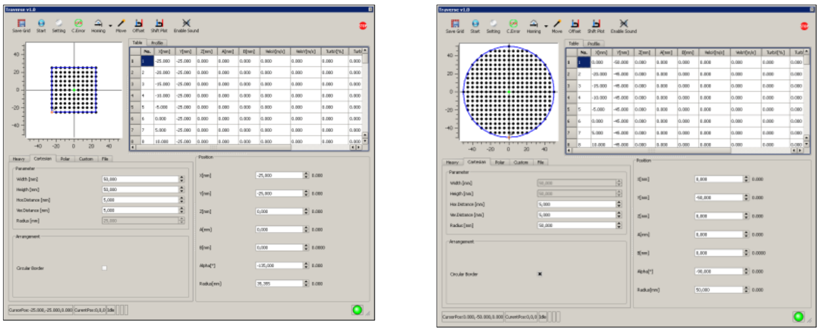


Figure 35: Cartesian Grid

8.7.3 Polar Grid

For applying a polar grid to the measuring-area use this option. It is basically the same procedure as described in chapter 8.7.1. Here you have the option to apply the grid in radial direction. Therefore parameters such as radius, layer and line have to be defined. You can choose between circular and linear arrangement. That defines how the grid will be scanned by the traversing unit (circular ➔ circumferential, linear ➔ radial).

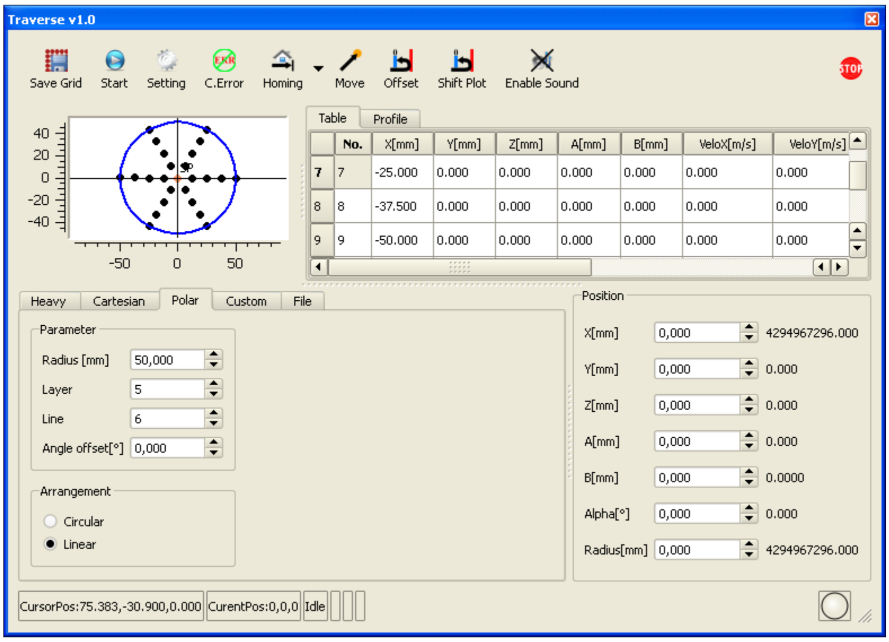


Figure 36: Polar Grid

8.7.4 Custom Grid

There is also the possibility to set your own grid in the custom tab. You will find for each axis (x and y) three buttons which allows to add, delete and refreshes your entries. The grid is created as a matric with x as row and y as column.

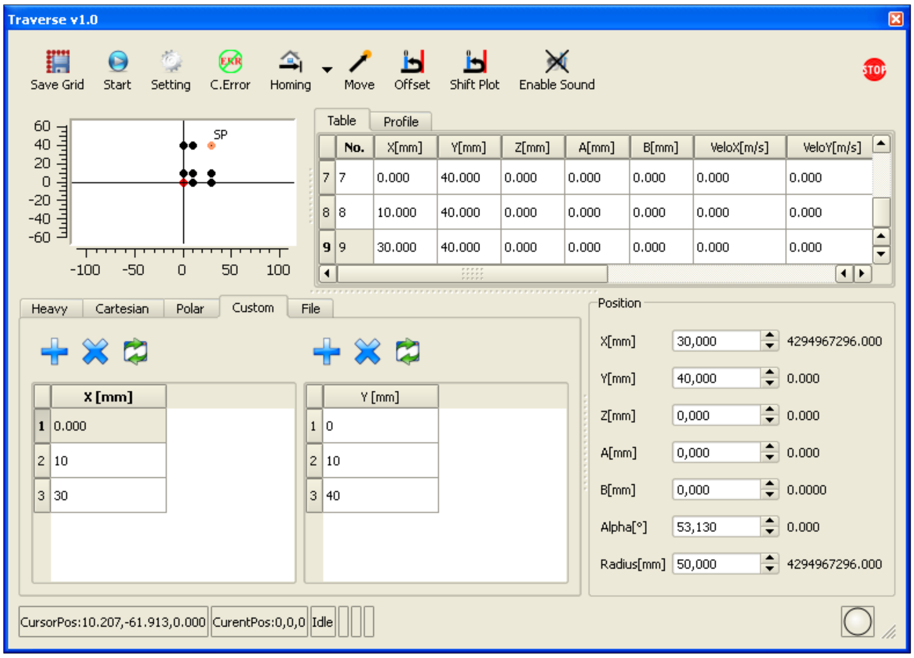


Figure 37: Custom Grid

8.7.5 File Window

This tab allows you to load any previous build grid in to traverse menu. The file-format that can be load is *.trm.

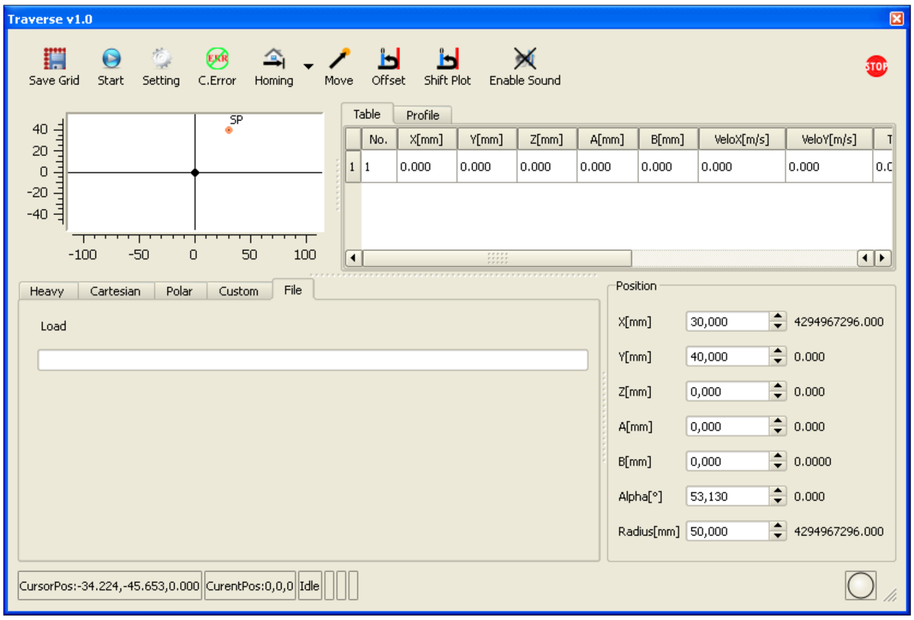


Figure 38: Load existing grid

8.8 Profile Presentation

The measurement system is able to plot the averaged velocities with respect to the measurement volume position.

Figure **39** depicts the velocity distribution along the x axis of a measurement grid with a length of 100 mm.

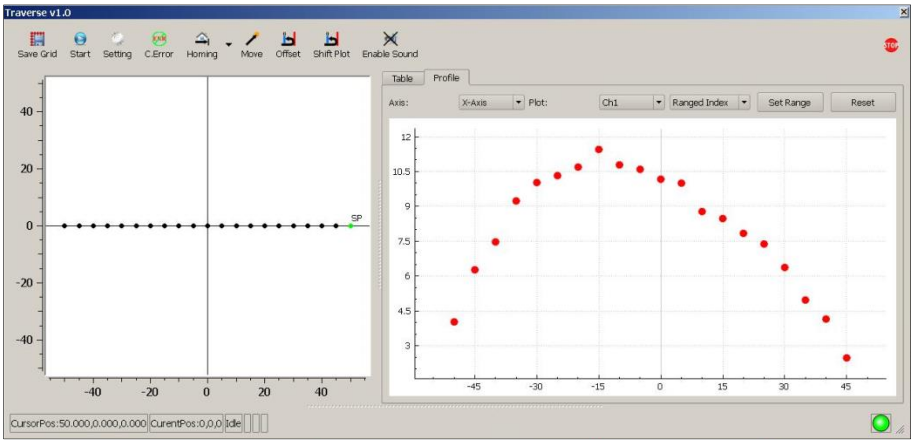


Figure 39: Measured Velocity Profile

The velocity distribution can be plotted with respect to any axis (x,y,z). If you want to view only a certain range of MPs. Choose ranged index to appropriate range and click on set range. That will update the velocity profile to the chosen index range of MPs. You also have the possibility to change the velocity value directly as a manual input in the tab called table shown beside the plot.

8.9 Storing data

The LDA Control QT software works as tool for the capture, processing and representation of transient, high frequency signals with a range of max. 400 MHz. The software works in combination with a very fast A/D converter card which allows sampling rates up to 400 MHz. In particular LDA Control QT is suitable for the analysis of Doppler burst signals in Laser-Doppler- Velocimetry (LDV/LDA) applications.

Laser Doppler Velocimetry is often used for making pointwise of velocity field measurements, so there is a need to store the velocity data in a way that relates it to measurement position. To simplify the measurement and data archiving process, LDA Control QT takes the following approach: A Master File (*.ldv) is opened by the user, in which the measured data for the individual measurement points are registered. To minimize the flow of data, only the most important data items are registered into the master file (measurement point index, measurement point coordinates, temporal average value of the velocity on channel 1, turbulence degree 1 and Doppler frequency 1, followed by the data of the second channel). See the example below (extracted from a LDV-file):

8.9.1 LDV-file

```
1 0.000 0.000 0.000 0.162686 33.861618 30081.875651 341 06/03/14 15:28:46 06/03/14 15:28:46
2 0.000 0.000 0.000 0.172350 19.258747 31868.851297 727 06/05/14 13:43:51 06/05/14 13:44:36
3 0.000 0.000 0.000 0.128141 31.070775 23694.303583 476 06/05/14 13:51:49 06/05/14 13:52:32
4 0.000 0.000 0.000 0.175738 24.704583 32495.245504 122 06/05/14 14:15:45 06/05/14 14:15:52
```

Figure 40: Format of the LDV-file for a 1D measurement without used analog inputs

In the shown order above (Figure 40), the columns correspond to the following quantities: current measuring point number, x-coordinate, y-coordinate, z-coordinate, velocity channel1, turbulence degree1, Doppler frequency1, number of valid bursts, the start date, the start time, the end date and the end time.

If the analog input option is used you will get additional two columns for each input channel (totally twelf for six input channels): the mean value and the standard deviation in percent (relating to the men value) about the measuring time. These columns are positioned between the column for the number of bursts and the column for the start date of the measuring point.

```
******************* Measurement Values *******************
Velocity[m/s]: 0.0075921 Turbulence[%]: 2.33248 Doppler Frequency[Hz]: 7592.100612 Turb.Freq[Hz]: 0 Prob.: 0
******************* Parameters ***************************
Samples : 16384 Burst: 100 Shift[Hz]: 0.000000 CalFactor: 0
Sample Rate[Hz] : 50000000 BValid: 2.50 BThres[mV]: 10.000 Fringe Distance[um]: 1.000
Position[mm]: 0 0 0
Start Time: 11:00:34 End Time: 11:00:36
******************* Data *********************************
Nr.   fd[Hz]     t[s]  v.LDV[m/s]
1 7452.109572 0.000000 0.007452
2 7596.633870 0.063000 0.007597
3 7762.905840 0.266000 0.007763
4 7301.352014 0.297000 0.007301
5 7826.864809 0.329000 0.007827
6 7449.514698 0.360000 0.007450
7 7774.350691 0.391000 0.007774
```

Figure 41: Format of the CH0 file

Additionally, for each measurement point a Data File *.ch0/*.ch1 (Histogram File) is produced in which the individual Doppler frequencies, the burst-valid criteria, the interference fringe spacing and the shift frequency are registered. In order, the columns correspond to the following quantities: measuring point number, Doppler frequency, arrival time in seconds, phase difference (if activated) and velocity. To make a simple interface to commercial programs and/or user programs, both master files and CH0/CH1 data files are stored in ASCII format.

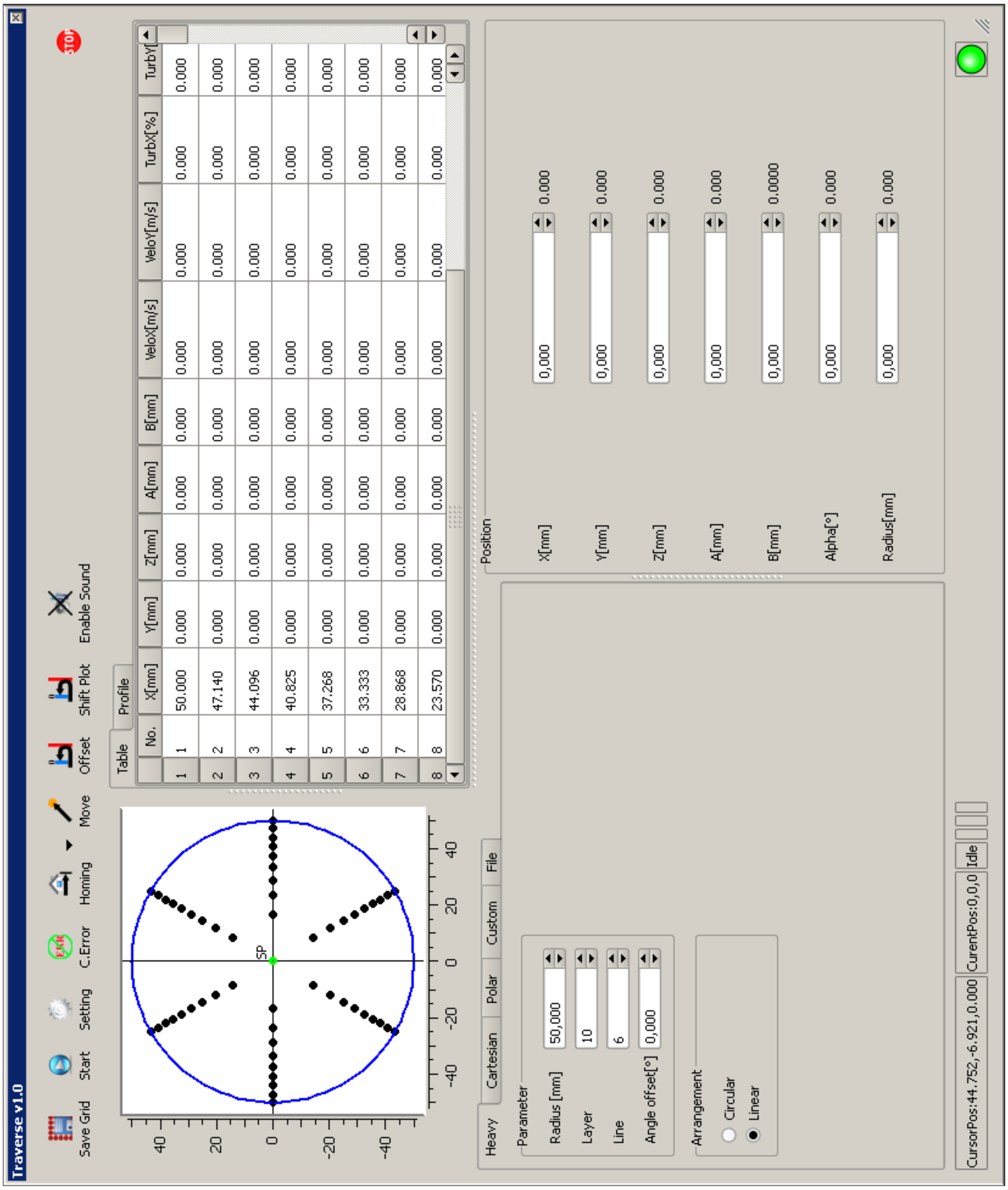
```
Nr.;fd[Hz];t[s];v.LDV[m/s]
1;7452.109572;0.000000;0.007452;
2;7596.633870;0.063000;0.007597;
3;7762.905840;0.266000;0.007763;
4;7301.352014;0.297000;0.007301;
5;7826.864809;0.329000;0.007827;
```

Figure 42: Format of the CSV file

Otherwise it exist a result file in the CSV EXCEL format for each measuring point (Figure 42).

9. Appendix

9.1 Heavy Grid



9.2 Rotated Heavy Grid